



FOOD FOR THOUGHT

Ryan Fernando



FOR MORE
STORIES

Root of the matter: The science of plant-powered performance

Plant-based eating can fuel sporting success, but only when **backed** by science, structure, and the right nutritional choices.

my two decades of working with Indian athletes — from Olympic medallists and cricket stars to weekend warriors chasing personal bests — one question keeps coming back to me with growing frequency: “Ryan, can I perform at the highest level on a plant-based diet?”

The short answer is yes. The honest answer is: ‘only if you build it right’.

India has a unique sporting context. Roughly four out of ten Indians grow up vegetarian, and a rising number of young athletes are now experimenting with veganism for ethical, environmental, or gut-health reasons.

Yet many still treat plant-based eating the way their grandmothers did, heavy on rice, rotis, dal, and a bit of sabzi. That may be fine for a desk job. It is nowhere close to what an athlete’s body demands.

Let me walk you through what actually works on the ground.

Start with protein, because that’s where most athletes fall short

The biggest gap I see in vegetarian and vegan athletes in India is protein. Not just quantity, but quality

and distribution.

An athlete training seriously needs roughly 1.6 to 2.2 grams of protein per kilogram of bodyweight every day. A 70-kg footballer or runner is looking at 112 to 154 grams daily. Most vegetarians I assess during their first consultation are eating barely 45 to 60 grams, and most of it lands in one heavy dinner. That pattern simply does not support muscle repair.

Spread your protein across four to five meals. Aim for 25 to 35 grams per meal. This is where you actually drive muscle protein synthesis.

For vegetarians, the strongest building blocks are paneer, curd, eggs (if you include them), Greek yogurt, milk, and whey protein. For pure vegans, you need to lean on soya chunks, tofu, tempeh, rajma, chana, moong, urad, masoor, peanuts, pumpkin seeds, hemp seeds, and a good-quality plant protein blend — ideally pea and rice combined, because that pairing covers the amino acid profile that single sources often miss.

A practical tip I give every plant-based athlete: combine cereals and pulses in the same meal. Rajma chawal, dal khichdi, idli sambar,

chana with roti. These traditional pairings are not just nostalgic. They are amino-acid science your great-grandmother understood by instinct.

The nutrients you will almost certainly run short on

Over the years, I have tested hundreds of vegetarian and vegan athletes for micronutrient status. The deficiencies repeat like clockwork.

Vitamin B12 is non-negotiable for vegans. It is not present in any plant food in usable amounts. Get yourself a methylcobalamin supplement, 500 to 1,000 mcg daily or as your sports physician advises, and recheck your levels every six months. Low B12 can quietly erode both endurance and nerve function.

Iron matters massively, especially for female athletes. Plant iron is non-heme (not sourced from animals) and less readily absorbed. Pair iron-rich foods such as spinach, beetroot, jaggery, dates, and ragi with Vitamin C sources like amla, lemon, guava, or capsicum in the same meal. Avoid tea and coffee for one hour before and after iron-rich meals. That single behaviour change can significantly improve absorption.